

COMP 10066

Week 11 Lecture

Website Testing Lecture #2

Website Testing – What?

- We previously discussed the following topics for what needs to be tested
 - Content – content is unique and correct, error free including spelling and grammar
 - Search Engine Optimized – website has keywords to ensure it shows in search engines when appropriate
 - Readability – is the content readable by the intended audience
 - Functionality – do all links, buttons, forms work

Website Testing – What?

- We will continue our discussion with:
 - Accessibility – Does it meet standards? Does it work regardless of device?
 - Usability – Is it “easy to use”?
 - Performance** – Website is responsive, can handle expected load
 - ** will be covered in Week 12

Website Testing

- We will now discuss each of these components that need to be tested investigating the following:
 - Is the testing subjective or objective?
 - What testing techniques can you think of?
 - Is it possible to automate this testing?

Accessibility

- List of reasons why making your application accessible to persons with disabilities
 - A possibility significant portion of the market
 - Morally the "Right thing to do"
 - Require by law
- Example: Accessibility in Ontario
 - Websites must be made accessible under AODA
 - <https://www.ontario.ca/page/how-make-websites-accessible>

Examples of Accessibility

- Visual
 - Allow (obey) user selection of color – where possible do not hardcode colors into the application
- Hearing Impaired
 - Program should provide data in other forms textual (closed caption)
- Motion impairment
 - Users may have difficulty using mouse / keyboard
 - Speech recognition software
- Cognitive Learning
 - Multiple language support can help
 - Simple terms where possible
 - Evaluate Content Level

Examples of Accessibility

- Can you think of more examples that pertain to website design?

Accessibility Testing

- Is this subjective or objective?
- What techniques can we use to ensure websites are accessible?
- Can this be automated?

Accessibility Testing

- Objective
 - Standards and Guidelines exist:
<https://www.w3.org/WAI/standards-guidelines/wcag/>
- Tools do exist to verify certain features for accessibility
- Tools may focus on a certain type of impairment
- By investing in Accessibility Testing, covering many aspects of usability as well

Usability

- “Easy to use” – cannot measure any requirement that is easy to use
- Requirements need to be measurable in nature, in order to write tests the correspond to usability
- Examples:
 - Number of clicks to accomplish a task
 - Percentage of failed tasks cannot exceed a certain amount
- What is the most annoying user interface “feature” you can think of from sites or apps you have used?

Usability Testing

- Is this subjective or objective?
- What techniques can we use to ensure websites are accessible?
- Can this be automated?

Usability Testing

- Subjective – often varies from user to user
- Market research and focus groups used to get feedback
- Follows best practices and traits of a good user interface
- Mostly manual

Traits of a Good User Interface

- Provide a list of important traits that are inherent in a good User Interface (UI)?
 - Follows Standard Guidelines
 - Intuitive & Useful
 - User should feel comfortable with user interface
 - Correct & Consistent
 - Forgiveness
 - Allows the user to recover from errors or unwanted actions
 - Protection
 - Stops the user from doing unwanted actions
 - Minimal use of “jargon”

How do we define best practices?

- Look to the organization
 - Windows – Microsoft
 - Mac / iPhone – Apple
 - Android – Google
 - Web – "Web Content Accessibility Guidelines"
 - <https://www.w3.org/WAI/standards-guidelines/#wcag>

What is included in those best practices?

- Interface Consistency
- Integration Between Applications
- User Content
- User Interaction / Control
- Feedback to User
- **Keep it Simple**

Usability - Devices

Daily hours spent with digital media in the United States

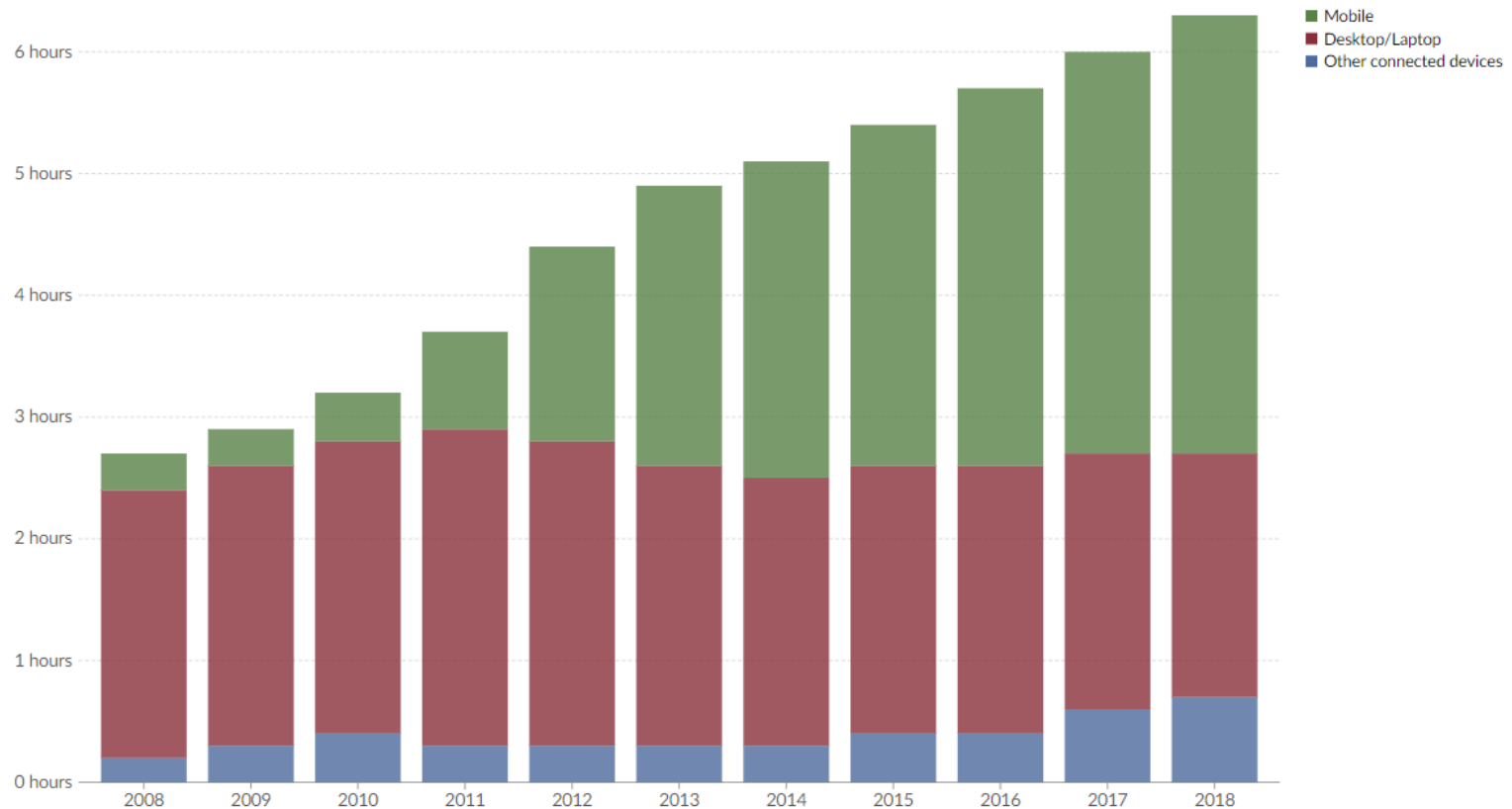
Average daily hours spent engaging with digital media (e.g., images and videos, web pages, social media apps, etc.) The data for 'other connected devices' includes game consoles. Mobile includes smartphones & tablets. All data includes usage at home and work for people 18+.

Our World
in Data

Table

Chart

Settings



Data source: BOND Internet Trends (2019) - [Learn more about this data](#)

OurWorldInData.org/internet | CC BY

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Configuration and Compatibility Testing

- What is Configuration and Compatibility testing?
 - Testing of site by using a combination of different hardware and software.
 - This type of testing is important in a fractured hardware/software environment
 - Current Web / Mobile device environment is challenging for testing
- **Name some of the challenges?**

Checklist of Considerations

- Hardware used
- Mobile vs Desktop vs Tablet
- Software / Operating System
- Browser used
- Speed of connection – cannot assume high speed
 - <https://crtc.gc.ca/eng/internet/internet.htm>
- Plug Ins

What to test?

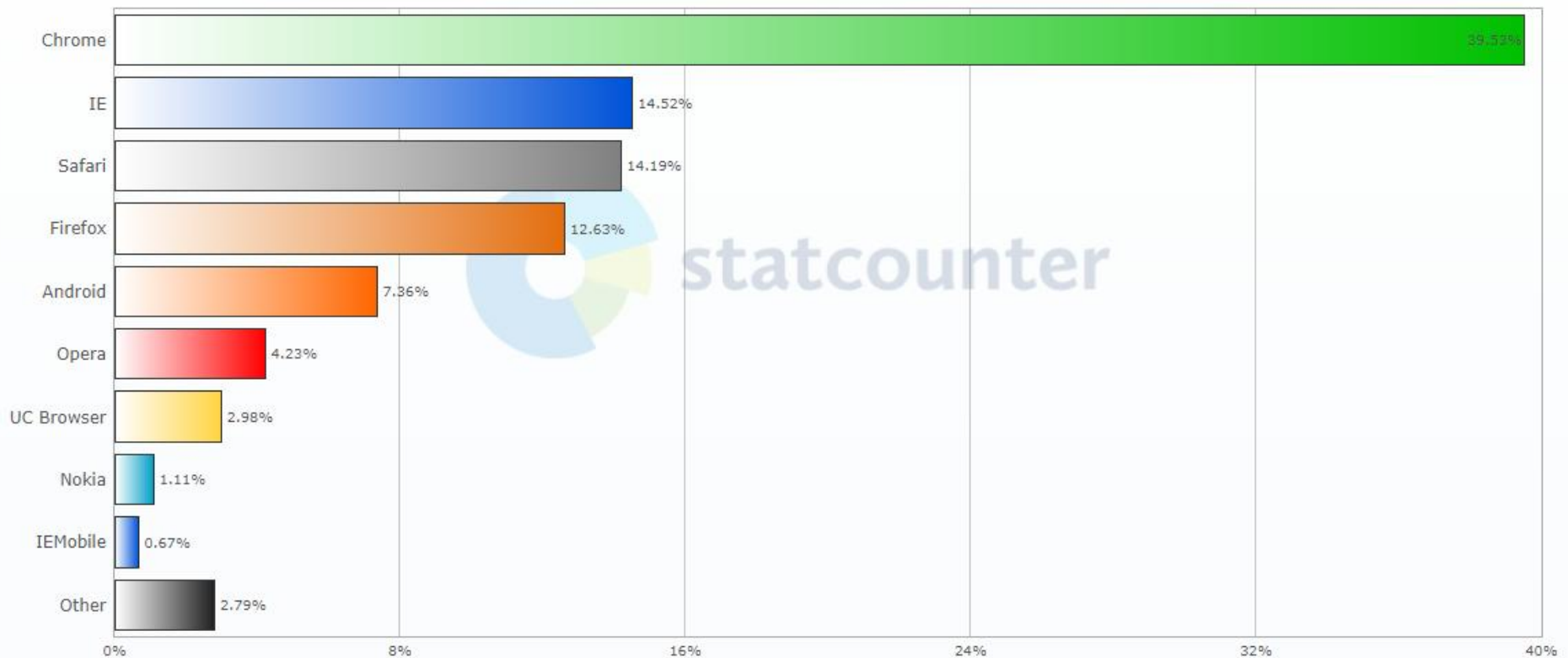
- Due to time constraints we cannot test every combination
- Expensive to maintain testing systems for many different configurations
- What do we choose then?

Top Browsers 2014

Browser Market Share Worldwide

Aug 2014

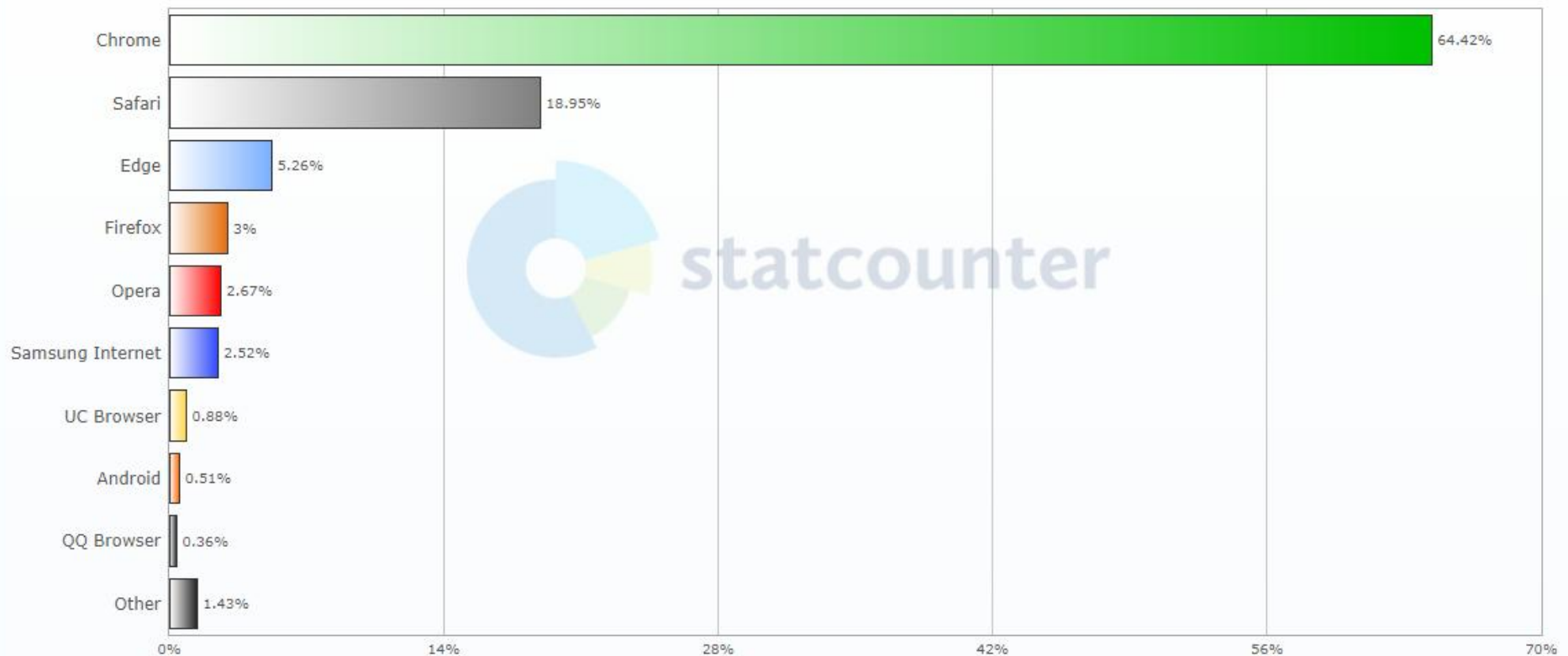
[Edit Chart Data](#)



Top Browsers 2024

Browser Market Share Worldwide

Aug 2023 - Aug 2024

[Edit Chart Data](#)

What do we choose?

- We choose the most popular or what we know the majority of our users will be using
- May run a small selection of “sanity” tests on additional configurations to know the system will function at the basic level

Automation Tools Help

- Tools like Selenium WebDriver allow the same Unit Tests to be run on multiple web browsers with small changes in configuration
- Test cases are set up once – time saving